

COURSE TITLE : PLANT UTILITIES AND MAINTENANCE
COURSE CODE : 6075
COURSE CATEGORY : E
PERIODS/ WEEK : 4
PERIODS/ SEMESTER : 60
CREDIT : 4

TIME SCHEDULE

MODULE	TOPIC	PERIODS
1	Plant Utilities – Water& Air	15
2	Steam and steam generators	15
3	Operational Maintenance of Chemical Plant equipments	15
4	Plant location and lay out	15
TOTAL		60

COURSE OUTCOMES :

SL.NO	SUB	STUDENT WILL BE ABLE TO
1	1	Appreciate the requirement of different utilities for the process plant.
	2	Know the generation and its effective utilization of water ,steam and air
	3	Understand different equipments used to run the process plant with different utilities.
2	1	Acquire the knowledge for selection of different utilities.
	2	Understand the cleaning methods of Tubular equipments.
	3	Comprehend the start up and commissioning of chemical plant equipments
3	1	Understand the factor to be considered for the selection of chemical plant.
	2	Understand the chemical plant ventilation and lighting

SPECIFIC OUTCOME

MODULE-I

UTILITY AIR, WATER, AND STEAM

1.1.0 Understand utility Air and water

- 1.1.1 Name the air handling Equipments
- 1.1.2 Preparation methods of Instrument air
- 1.1.3 Discuss about heating and cooling of air

1.2.0 Comprehend Importance of water in chemical industries and water purification process

- 1.2.1 List the source of water
- 1.2.2 List the characteristics of water
- 1.2.3 Define hardness
- 1.2.4 Define temporary and permanent hardness
- 1.2.5 State the disadvantages of hard water
- 1.2.6 State the scale and sludge formation in boilers

- 1.2.7 List the disadvantages of scale formation
- 1.2.8 Define caustic embrittlement
- 1.2.9 Explain the following softening methods 1)lime soda process 2) zeolite process.
- 1.2.10 List advantages disadvantages cold and hot lime soda process
- 1.2.11 Explain the ion exchange process
- 1.2.12 Explain the method of municipal water treatment.
- 1.2.13 Describe the Production of cooling Water and chilled water
- 1.2.14 Explain the methods of distribution of cooling water
- 1.2.15 Describe preparation of Boiler Feed Water.

MODULE- II

Steam & Steam Generators

2.1.0 Understand the various properties of steam

- 2.1.1 Explain the formation of steam under constant pressure.
- 2.1.2 State the effect of pressure on temperature.
- 2.1.3 Distinguish the wet steam, dry steam and super heated steam.
- 2.1.4 Explain the properties of steam such as sensible heat, latent heat, total heat of steam, superheat and dryness fractions.

2.2.0 Understand the working of steam Generators

- 2.2.1 Define a boiler.
- 2.2.2 Explain the functions of a boiler.
- 2.2.3 Distinguish between fire tube boiler and water tube boiler.
- 2.2.4 Distinguish between low pressure boiler and high pressure boiler.
- 2.2.5 Describe with line diagram the construction and working of a Simple Vertical Boiler and Cochran boiler, Babcock and Wilcox boiler, la -Mont Boiler and Benson boiler.
- 2.2.6 Compare the smoke tube and water tube Boiler.
- 2.2.7 List the specification of a boiler.
- 2.2.8 State the function of boiler mountings such as stop valve, safety valve, water level indicator, pressure gauge and fusible plug etc. with line sketches.
- 2.2.9 Explain the functions of boiler accessories such as economizer, feed pump, super heater and air pre-heater etc. with simple sketch
- 2.2.10 Explain the developments such as draft, natural draft, induced draft, forced draft and balanced draft.

MODULE – III

Operational Maintenance of Chemical Plant equipments

3.1.0 Understand the principles of maintenance management

- 3.1.1 List the need for maintenance.
- 3.1.2 List the three types of maintenance.
- 3.1.3 Prepare a sample maintenance schedule for any equipment.
- 3.1.4 Describe the cost factors in maintenance.
- 3.1.5 List the main elements in a maintenance records.
- 3.1.6 List the tree types of equipment replacement.

3.2.0 Comprehend maintenance of Chemical Plant equipments

- 3.2.1 List the methods of cleaning tubular equipments.
- 3.2.2 State why cleaning of tubular equipment is essential.
- 3.2.3 Explain method and procedures of chemical cleaning of tubular equipment.
- 3.2.4 Describe the mechanical methods of cleaning tubular equipments.
- 3.2.5 List the method for cleaning towers and columns.
- 3.2.6 Explain the common causes of troubles and remedial action in the following equipments: Centrifugal pumps, Reciprocating compressor, Filter press, Agitator
- 3.2.7 Explain the decoking methods of furnace tubes.

3.3.0 Know start up and commissioning of some plant equipments

- 3.3.1 Describe the procedure for starting of centrifugal pumps.
- 3.3.2 List the precautions to be taken before starting the reciprocating pump.
- 3.3.3 List the precautions to be taken before starting of the reciprocating compressor.
- 3.3.4 Explain various steps to be taken for the start up and commissioning of the heater.
- 3.3.5 Explain various steps to be taken for start up and commissioning of a distillation column.
- 3.3.6 Select various steps to be taken for start up and commissioning of the Absorption columns.

3.4.0 Understand chemical plant inspection

- 3.4.1 Explain the precautions to be taken before opening and entry in the vessels for cleaning.
- 3.4.2 List the types of inspections.
- 3.4.3 Explain conventional monitoring methods.
- 3.4.4 List the standard testing methods of a high pressure vessels before and after fabrication.
- 3.4.5 List the non-destructive testing methods.
- 3.4.6 Describe the ultrasonic test for pressure vessel.
- 3.4.7 Describe the radiographic test for pressure vessel.

MODULE – IV

Plant Location and Layout.

4.1.0 Know the basic principles in location and layout of a chemical plant.

- 4.1.1 List the points to be considered in the location of a big chemical plant.
- 4.1.2 Identify the role of raw materials, transportation and power supply in the location of a Chemical plant.
- 4.1.3 Explain the community factors in the location of a chemical plant.
- 4.1.4 State the waste disposal and water supply in the location of a chemical plant.
- 4.1.5 Explain the factors to be considered in the preparations of a plant lay out.
- 4.1.6 Draw a plant lay out for a chemical plant.
- 4.1.7 List the safety factors involved in plant layout.
- 4.1.8 Locate a canteen, ETP, Boiler house, in a typical layout and justify the location.
- 4.1.9 List the methods of preparing a plant layout and explain any one method.
- 4.1.10 Explain master plot plan and unit plot plan.
- 4.1.11 Draw a typical master plot plan of a plant you know.
- 4.1.12 Explain the guide lines for preparing a master plot plan and unit plot plan
- 4.1.13 Illustrate the scale model method of preparing plot plans.

- 4.1.14 Draw sketches of different types of chemical plant buildings.
- 4.1.15 Differentiate between housed and Un housed plants.
- 4.1.16 Explain the advantages of multi storied construction over single storied construction.
- 4.1.17 Explain chemical plant roofs and floorings.
- 4.1.18 Describe the methods of providing artificial and natural ventilation and lighting.

COURSE CONTENT

MODULE I

UTILITY- AIR AND WATER

Cooling Water - Production of Cooling Water – Distribution of Cooling Water – Chilled Water – Preparation of Chilled Water – Boiler Feed Water – Water treatment methods - Lime soda process – Hot and cold - Advantages – Disadvantages – Ion exchange Process - Municipal water treatment Fire Fighting Water – Air handling Equipments, Instrument Air – Preparation methods – Heating and Cooling of air – Compressed Air - Plant air —

MODULE II

PROPERTIES OF STEAM & BOILERS

Properties of steam – Explanation of properties of steam such as total heat of water - latent heat- total heat of steam- super heat - dryness fraction. Function of boilers – classification – working principle – fire tube and water tube boilers – Simple vertical boiler – Cochran boiler- Babcock and Wilcox boiler High pressure boilers – Benson boiler – Lamont boiler - Boiler mountings – accessories.

MODULE III

OPERATIONAL MAINTENANCE OF CHEMICAL PLANT E Q U I P M E N T S

Principle of management – types of maintenance - Breakdown maintenance - scheduled maintenance - Preventive maintenance – Advantages - Objective of Preventive maintenance - schedule – records – replacement – maintenance of chemical plant equipments – Troubles and remedial actions – start up and commissioning of equipments – plant inspection – vessel entry – testing methods– non destructive testing methods – radio graphic & ultrasonic test.

MODULE IV

PLANT LOCATION AND LAY-OUT

Location of Chemical Plant – Important Factors considering Plant Location and Plant Lay-out – Master Plot Plan and Unit Plot Plan – Chemical Plant -Buildings – Roofing – Flooring – Ventilation and Illuminations.

REFERENCE

- | | | |
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| Prof.BENCHAMIN | - | General Engineering |
| BADGER & JULIOUS T BANCHERO | - | Introduction to Chemical Engg. |